Filling clinical coverage gaps with medical knowledge

BKO status update

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Question: can medical knowledge, held as a Bayesian Knowledge-driven Ontology, find coverage gaps in clinical data and generate synthetic records that improve prediction for underrepresented groups?

Glossary

Term	Meaning	Term	Meaning
BKO	Bayesian Knowledge-driven Ontology: ontologies fused with the data	CTGAN	Conditional Tabular GAN: neural table generator
BKB	Bayesian Knowledge Base: the BKO's probabilistic store	TVAE	Tabular Variational Autoencoder: neural table generator
MONDO	Mondo Disease Ontology (diseases)	GAN	Generative Adversarial Network
HPO	Human Phenotype Ontology (findings)	SMOTE	Synthetic Minority Oversampling: interpolates real rows
ICD-9	Intl. Classification of Diseases, 9th rev.	ADASYN	Adaptive Synthetic sampling (SMOTE variant)
LLM	Large language model (Haiku 5.5)	CART	Classification and Regression Trees
CLLM	Curated LLM: LLM shown real rows writes more	Copula	Gaussian copula: columns + correlations
LR	Logistic regression (criteria model)	BN	Bayesian network: learned column graph
HGB	Histogram gradient boosting (trees)	DCR	Distance to closest record (privacy)
Recall	Share of true positives caught	AUC-PR	Area under precision-recall curve
Holdout	One train/test split; we average 100	E1-E6, G6, G7	Our experiment IDs

How BKO works

1. **Represent:** fuse MONDO and HPO with the dataset into a BKB, weighting each statement by how often the data shows it.
2. **Detect:** find groups (age band × sex × label) whose own records leave medically expected patterns unsupported.
3. **Generate:** fill each gap group with new records, either by
   - an **LLM** (`bko`), prompted with the group, its gaps and summary rates, never an individual record; or
   - **direct draws from the BKB** (`bko-direct`): no LLM, no cost.
4. **Validate:** keep only records that pass consistency and range checks.
5. **Evaluate:** add the records to the training data and score recall and AUC-PR on held-out patients, beside every comparison method.

Every other method learns from the training rows alone. BKO also brings ontology structure, so it can fill a group that has **no rows at all**.

Results so far

Experiment	Setting	BKO result
E3 · Synthea	A subgroup's cases removed	Recall on the removed group 0.262 → **0.400** (+0.138 [+0.038, +0.237]); level with TVAE 0.403 and CTGAN 0.393; **leads under HGB** (0.311 vs CTGAN 0.251)
E1, E4, E5 · UCI Heart Disease	No removed group	Roughly **neutral**: the data already covers its groups
G6 · generator model	Which LLM writes rows	Sonnet 5.5 rows score higher than Haiku 5.5 (+0.066); local models lower; Haiku is 11× cheaper per record
G7 · likelihood curation	Keep the BKB's likeliest rows	Did not separate on UCI; to be retested on Diabetes
CLLM pilot · Diabetes	Curated CLLM	Kept **0 of 46,360** records; dropped (saved about $1,700)

E6: a removal test on real hospital data

Data: UCI Diabetes 130 hospitals, 69,283 adult first encounters, label = readmitted within 30 days (9.0%).

Target group: men aged 60-79 who were readmitted. Two conditions, 100 holdouts each, 16 methods:

- **98.5% removal:** 19 true rows stay. Does knowledge help when only a few cases were collected? *(main reading, due tonight)*
- **100% removal:** no true rows stay. Can knowledge stand in for cases never collected? **Complete.**

What 100% models: a coverage gap, not bad data. Examples:

- readmissions recorded at another hospital;
- outcomes lost to follow-up;
- small cells suppressed for privacy;
- a model moved to a new population.

E6 at 100% removal: BKO is the best method

Recall on the removed group (mean over 100 holdouts). `real` puts the true rows back: the ceiling.

Method	Recall (LR)	AUC-PR (LR)	Recall (HGB)
`real` (ceiling)	0.556	0.156	0.585
`bko-direct`	**0.538**	**0.152**	**0.359**
CTGAN	0.478	0.136	0.274
Gaussian copula	0.351	0.144	0.045
`bko` (LLM)	0.308	0.125	0.045
CLLM	0.282	0.131	0.087
none, SMOTE, ADASYN, TVAE, CART, BN	0.066	0.153	0.000

- `bko-direct` recovers **97% of the true rows' recall**. It beats CTGAN by +0.059 [+0.011, +0.108], with **no AUC-PR cost** on the group.
- **Knowledge beats examples:** both BKO arms beat CLLM, the same LLM shown real rows.
- Methods that copy rows have nothing to copy, so they add nothing.

How `bko-direct` builds a record

1. **Concept rates:** for each diagnosis and lab concept, `share = (n_cell·p_cell + 20·p_group) / (n_cell + 20)`. With the group empty, this is the readmitted-patient rate.
2. **Concepts:** draw each concept present or absent independently, at its share.
3. **Fields:**
   - sex and label are set to the target group's;
   - age comes from a readmitted patient aged 60-79;
   - each concept field matches its drawn concept;
   - every other field is copied **independently** from a different random readmitted patient.
4. **Validate:** all 126,900 rows passed.

Why it works: LR adds up feature effects, so real values from the right patients move every feature the right way. **Its limit:** independent draws break correlations between fields, so HGB gains less (0.359 vs 0.585). **Its assumption:** the missing group resembles other patients with the same outcome.

Privacy

What leaves the machine

- **CLLM** sends 20 real patient rows per API call.
- **`bko`** sends aggregates and ontology statements only.
- **`bko-direct`**, CTGAN and the classical methods run fully locally.

DCR, 5th percentile (100% removal, mean of 100 holdouts; higher is more private)

Real test patients (baseline)	`real`	Copula	`bko`	`bko-direct`	CLLM	CTGAN
0.0031	0.0125	0.0229	0.0211	**0.0194**	0.0171	0.0164

- Every generator's closest rows sit **5-7× farther** from training patients than real patients sit from each other.
- No exact test copies, apart from one row from `bko-frequency`.
- `bko-direct` is farther than CTGAN (+0.0030 [+0.0026, +0.0035]) and closer than the copula.
- **Caveat:** each value `bko-direct` writes is a real patient's value, though each field comes from a different patient. No method here offers formal differential privacy.

Why BKO, even at equal accuracy

- **Works with no data:** it fills a group that has no rows, where SMOTE, ADASYN, TVAE, CART and the BN add nothing.
- **No record-level disclosure:** the LLM path sends aggregates only, and `bko-direct` runs on-device.
- **Finds the gaps:** E2 recovered the removed group's gaps, giving a data team an audit it can act on.
- **Explainable and replayable:** each row traces to a gap, its concepts and a logged draw or prompt.
- **Cheap:** $0 for `bko-direct`, $0.000136 per record with Haiku. No generator training.

Open caveats

- The win comes from the BKB's structure, not the LLM: `bko` trails `bko-direct` by 0.23 and CTGAN by 0.17.
- Most row-adding methods lower AUC-PR on the group; `bko-direct` does not.
- UCI was neutral: the value is specific to coverage gaps.

Status and next steps

E6 run:

- 100% removal: done, $198.80.
- 98.5% removal: about 60 of 100 holdouts done, due about 20:00-22:30 EST tonight.
- Projected total about $475, against a $1,500 cap.

Next:

1. **E6 close-out:** replay one slice per condition, merge, read the 98.5% results, record the evidence, then PR #5.
2. **Likelihood gate on Diabetes** (free). A curated BKO arm runs only if the gate separates (cap $80).
3. **New task, `privacy-identifier-perturbation`:** an LLM rewrites `bko-direct`'s identifying fields while keeping concepts and distributions. DCR is tracked per method.
4. **Candidates:** a faster BN draw, a full thinking-on local run, and closing the gap between `bko` and `bko-direct`.

Merged this batch: #69 G6 replay · #70 thinking cost · #71 Diabetes 130 frame · #72 E6 harness readiness